

Designing an Algal Relay Computer: A Critical Orientation in Exploring More-than-Human Temporality

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Figure 1: The prototype of ARC – Algal Relay Computer.

ABSTRACT

The re-evaluation of our industrial, anthropocentric relationships with time has led to the emergence of new conceptualizations of temporality beyond clock time in design-oriented HCI research. In order to provoke discussions on how design may (or may not) account for temporality of nonhumans, we conceptualized and prototyped Algal Relay Computer (ARC), a "slow" computer that incorporates the growth of algae in its calculation process. The paper

presents theoretical groundings and technical details of ARC and offers preliminary insights into multiplicity of more-than-human temporality uncovered through the design process and exhibition of this artefact.

CCS CONCEPTS

• **Human-centered computing** → *Interaction paradigms*.

KEYWORDS

temporality, more-than-human design, critical design, micro-algae, hardware

ACM Reference Format:

Yuta Ikeya and Bahareh Barati. 2023. Designing an Algal Relay Computer: A Critical Orientation in Exploring More-than-Human Temporality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing*

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585874>

Systems (CHI EA '23), April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3585874>

1 INTRODUCTION

As technologies continue to evolve, our interactions with them have become increasingly fast-paced, with an emphasis on optimizing performance and productivity. Built on such capitalistic, industrial values, modern human society has advanced restless industrial activities and caused numerous environmental problems, as well as the extinction of many other species.

Recent research in human-computer interactions (HCI) has begun to focus on new conceptualization of time and temporality [27], through which the researchers aim to move away from human-centric design approaches and to broaden our understanding of the ways in which technology and the environment are interconnected. This critical stance towards Anthropocene aligns well with so-called posthuman or more-than-human design that aims to reconnect us with nature. Engaging this topic situates designers within various temporalities such as the life cycles and reproductions of the nonhuman organisms which may occur at different paces than those of the designers. Despite this opportunity, concrete design cases that research and explore broader concepts of time and temporality beyond our perception through the use of tangible prototypes are currently limited.

We have designed *ARC — Algal Relay Computer*, a simple form of computer that incorporates the growth of micro-algae in its calculation process. ARC serves as a design provocation [3] conversation starter to engage design research community in the discussions concerning how we perceive time and how it affects our interaction designs with nonhuman entities. This project is centered around the question: How can designers incorporate more-than-human temporalities in their design practices to foster deeper exploration of human-nonhuman interaction design? It is important to note that this project is not meant to provide a comprehensive solution to such a complex, open question, but rather to serve as an exploratory step towards a better understanding of the more-than-human temporalities.

2 RELATED WORK

2.1 Temporality in HCI research and design

Deeply connected to space and material, i.e. materiality, time and temporality have been an important aspect in HCI research and design that influences our experience of technology and the design of systems and interfaces [23]. Various attitudes and interpretations of time have been considered throughout the history of HCI research.

According to the study by Wiberg and Stolterman [27], the first wave of the temporality studies in the field has focused on optimizing time and increasing efficiency through the use of technology. This approach viewed time as a scarce resource that needed to be managed and optimized. The second wave has been empirical studies involving designs built around temporality as a primary focus. These studies explored the ways in which design choices influenced our experience of time and how different approaches to temporality could be used to create more engaging technology. In the most recent third wave of research, a strand of HCI researchers

has been exploring new conceptualization of time and temporality beyond our conventional understanding of clock time. This includes looking at alternative ways of understanding and experiencing time, such as through cultural or philosophical perspectives. This approach reflects a growing recognition of the limitations of an anthropocentric approach to HCI and design, and an interest in considering the broader context and implications of technology on society and the environment.

Pschetz and Bastian [19] have proposed *Temporal Design*; a perspective to regard time as pluralistic instead of “uniform, external to practices, and in a state of continuous acceleration”. They promote the subjective experience of time that is “emerging out of relations between cultural, social, economic and political forces” (p. 169). Rahm-Skågeby and Rahm [20] have presented the possibilities of relating *deep time*, a concept concerning geological tempo-materialities (used in the fields of media archaeology, critical posthumanism, and the environmental humanities), in HCI research and design to develop new sensibilities. The authors claim that this deep time design thinking would be, for example, to be sensible of the temporal gap between short-term interactions visible to us and their consequences influencing the environment, and to go beyond the techno-solutionist perspective on technology and design.

Next to such conceptualizations and frameworks, a number of design researchers have attempted to challenge the complexities of temporalities through their tangible designs and practices. Often called *Slow Design* or *Slow Technology*, these works aim to invite a moment for mindfulness, self-reflection, critical reflection on technology, and so on [10, 16]. Hallnäs and Redström [10] have introduced the design agenda *Slow Technology* where the designed artefacts matter to the users beyond the moment it is actually being used. Their *Informative Art* [21] explores an aspect of wall paintings and pictures that communicates to visitors what kind of home it is.

Odom et al. [16] developed *Olly* and *Slow Game* to explore slowness in interaction design. *Olly* is a music player that plays music from the user’s listening history on Last.FM. It is placed in a room and occasionally chooses the track to be played independently of the user. *Slow Game* is a cubic-shaped device on which the user can play the game ‘Snake’. The game advances only one pixel every 18 hours and what the user does is flip the device to determine the next move. Through designing these research products [17], these researchers reflected on their choice of forms, materials, and computational qualities. However, as they emphasise in the article, designed artefacts and design practices addressing such forms of temporality are lacking in the field, and more research and practice initiatives to conceptualize diverse expressions of time are needed.

2.2 More-than-human design and temporality

More-than-human design (or posthuman design), is a growing topic in HCI research and design. This research topic challenges the human-centredness in design practices that have separated us from nature (among many other causes). Bastian[2] especially proposes that our insensibility to natural temporalities causes distancing between humans and nonhumans. According to such a proposition, the practice of attuning to more-than-human temporalities can help us reconnect to nature and improve our understanding of the

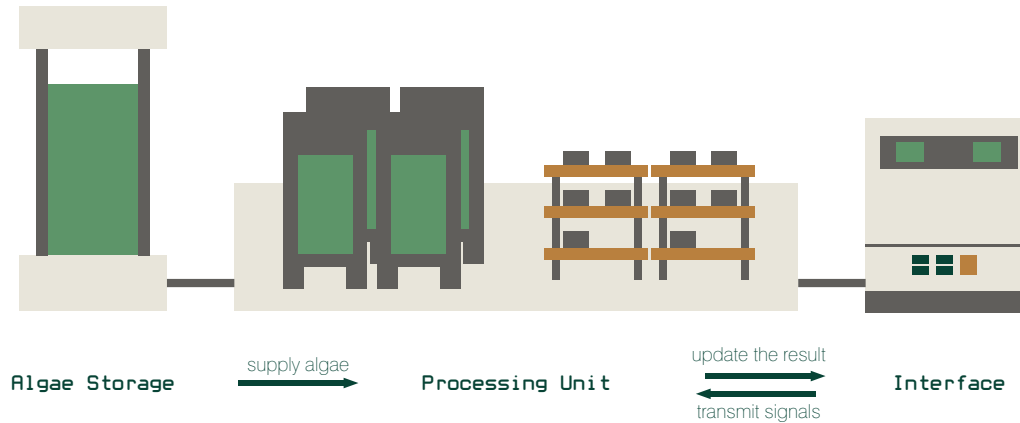


Figure 2: Components of ARC and the interactions between them: Algae storage; processing unit; and interface.

world around us [2]. The fields of bio-art and bio-design that often use “living artefacts” [13] or artefacts that incorporate nonhuman beings and showcases their livingness (e.g., [1, 7, 9]). However, there is still a need for more in-depth discussions and design research focused specifically on a more-than-human design context. While there have been some artistic explorations that have focused on time for other species and illuminated our narrow perception of time (e.g., [4, 12]), these examples often lack the rigour of academic research.

Considering the difference between humans and nonhumans in the lifespan and experience of time [11], it should be natural to consider temporality in more-than-human design. Nonetheless, the time and temporality aspect has been scarcely explored within this field through tangible artefacts. Below, we discuss two recent works that touch upon more-than-human temporality.

Ooms et al. [18] have designed a toolkit to prepare mud for the light installation [24]. More-than-human temporality in this project serves as an analytical lens, and the artefact is not necessarily designed through the temporal perspective. The work of Ciobanu and Juhlin [5] explores natural temporality and contrasts it with human clock time through the design of an internet browser that changes appearance depending on the brightness, temperature, and humidity of a specific location outside. The authors discuss how the experience of change in time differed through the technology-mediated prototype and through being out in nature without technology.

The discussion of time and temporality as a key component of more-than-human design can benefit from a larger repertoire of design cases. This is particularly crucial in the light of our limitations in understanding nonhumans [14, 25, 26]. As a consequence, the interaction designs that are thought and formed by the designers (humans) may be constrained by our perception of time. Further research in this area could contribute to the accumulation of knowledge on the incorporation of nonhuman temporalities into design and how this can impact our interactions with technology and the world around us.

3 METHODS AND DESIGN PROCESS

This project takes the showroom approach of constructive design research [15]. The designed artefact aims to disrupt the norms of everyday life and notion of temporality that are often taken for granted within design practice.

3.1 Prototyping ARC – Algal Relay Computer

ARC – Algal Relay Computer is a design installation of a computer that incorporates the growth of micro-algae (*Spirulina platensis*) in its calculation process. In contrast to usual computers we use everyday which returns the result of calculation within a fraction of a second, ARC spends substantially longer time such as a week (depending on the initial condition of the algae and environment) to derive a solution to a simple calculation. As a showroom artefact to evoke discussions, the extra attentions were paid to the aesthetics of the prototype (see Figure 1).

With the aim of ironically contrasting the computer technology: the symbol of instant response and the slowness of algal growth, we have assembled a prototype that achieves the desired function.

The ARC prototype consists of three components: the interface, the processing unit, and the algae storage. The interaction between the components are shown in Figure 2.

3.1.1 Engineering of the slow computer. The slowness of the ARC prototype is achieved by the relay consisting of algae culture and a simple electric circuit. Figure 3 shows the schematic of this algal relay. The algae culture in the container exposed to white, red, and blue LEDs (see Figure 4) will grow and increase its density [22] and decreases the amount of light that permeates through it [6]. A photoresistor is installed on the top of the algae container and coupled to the circuit that changes its state depending on the value of the photoresistor. In the initial stage, when the photoresistor receives an adequate amount of light, its resistance is low enough to allow only a minimal flow of current in the right-bottom part of the circuit (OFF). As the culture grows and the light reaching the photoresistor decreases, the resistance increases, resulting in a higher flow of current in the right-bottom part of the circuit (ON).

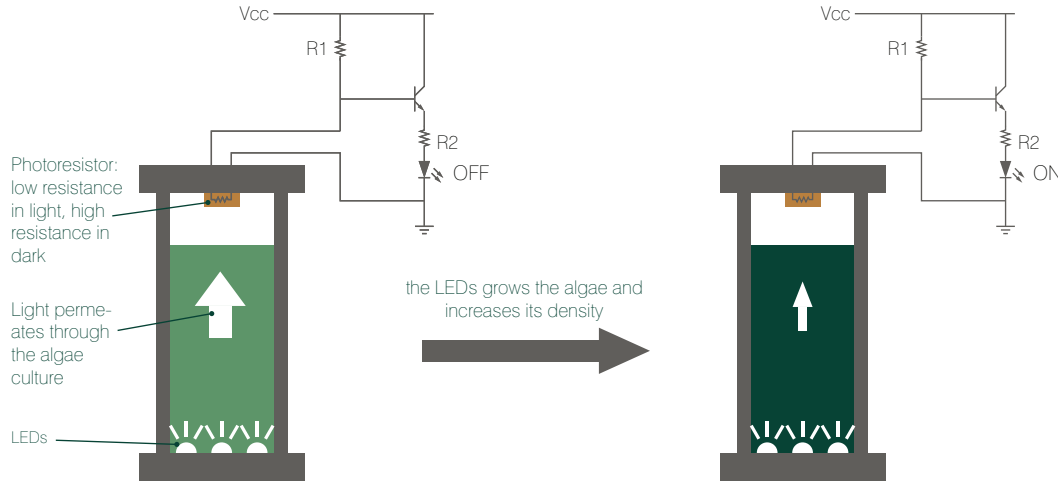


Figure 3: The switching principle of an algal relay.

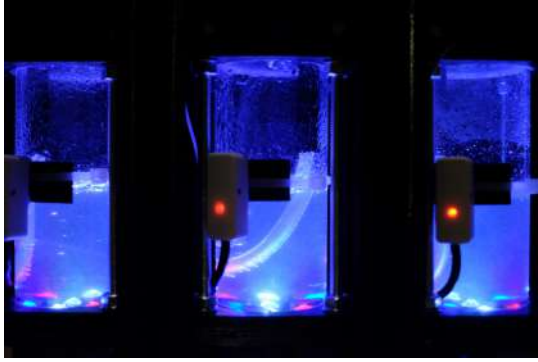


Figure 4: The picture of three algal relays on the prototype.

This switching threshold of density can be controlled by selecting the resistor R1.

3.1.2 Computation. A simple form of computer can be broken down to the combination of input, processing, output, and storage. The ARC prototype implements this algal relay between the input and the process to makes the total duration of calculation longer (see Figure 5). The storage function has been omitted from the prototype.

The prototype is capable of binary 2-bit calculations (addition) as a minimum representation of computer function. This minimization is due to the time and budget restriction of the project, however, it unintentionally emphasized the discrepancy between the simplicity of the calculation and the time required to complete it. To achieve this calculation, the processor of the ARC was built without any advanced micro-controller such as Arduino. Instead, transistors and relays are used for the logic gates and other switching functions within the circuit, respectively.

To perform the calculation $A+B$, a human inputs the numbers A and B in binary on the interface by pressing down the corresponding keys. When the return key is pressed, the valves in the processing

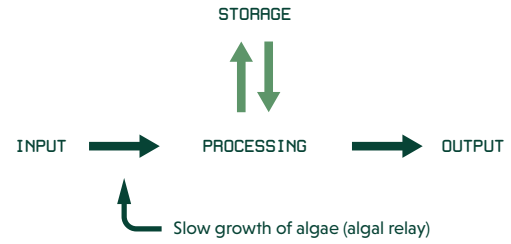


Figure 5: The simple schematic of functions on a computer. On ARC, the algal relays are implemented between the input and processing. Note that ARC does not have any data storage function.

unit open, and the algae culture is supplied from the storage to five algal relays. Four out of these five algal relays are assigned to the keys. The LEDs on the relays where the human assigned as HIGH are lit to grow the culture inside. The other relays that are assigned as LOW remain dark. After a certain time T_C (T_C should be as long as 30 minutes to a few hours), all the relays light up for a short period T_M (T_M should be as short as 1 to 5 minutes). During this T_M , the culture densities in all the relays are measured and the calculation is performed. The result of the calculation at this moment is displayed on the interface. As this cycle of $T_C \rightarrow T_M \rightarrow T_C \rightarrow T_M \rightarrow$ continues, the culture in the relays assigned to HIGH should grow as they are always exposed to light. Eventually, the relays that reach the threshold turns on and the result is updated every cycle. The cycle continues until the reference relay (the one that is not assigned to any key) is turned on. To achieve this order of switching, the switching threshold of the reference relay should be higher than the other four. After the reference relay turns on, the final result is continuously displayed. The flowchart of how this prototype performs as well as the entire electric circuit are available as supplemental materials.



Figure 6: Aesthetic details of the ARC prototype.

3.1.3 Aesthetic details of ARC. Since the ARC prototype serves such a limited function; 2-bit binary addition, we have chosen the aesthetic after early computers from 1970s that have relatively limited functionality compared to modern computers yet familiar to our eyes so that the viewers know that it is a computer. Accordingly, beige, the most common colour for the computers in 1970s and 80s, was chosen for the body colour of the prototype. We have avoided the overly “artistic” or “experimental” aesthetics that are seen in many algae-related works where many tubes are connected to transparent chambers (seen in e.g., [7, 8]). This is not to let ARC consumed as merely an art installation which is out of the world.

We chose to display the result of binary calculations as binary. The display has three rectangular apertures that can be illuminated independently to express the result of 2-bit binary calculation. The illuminated state of each aperture represents a value of 1, while the dark state indicates a value of 0. See Figure 6 for the details of the prototype.

3.2 Exhibition and reactions

Due to an unexpected leakage prior to the exhibition day, the algae culture could not be fed to the processing unit. By changing a few wiring in the processing unit, we managed to get the interface display 1-0-1 (as shown in Figure 6) every few minutes ($= T_C$) for a duration of 30 seconds ($= T_M$). This could mitigate the technical problem so that the prototype could seemingly function during the exhibition. The selection of such short cycle compared to our design intention (explained in 3.1.2) is motivated by the fact that the visitors would not stay at the booth more than five minutes most of the time.

The ARC prototype was exhibited for a day at Demoday at Eindhoven University of Technology hosted by Industrial Design Faculty. A diverse range of people among whom design researchers and students visited the exhibition. The prototype accompanied a short description in text. The first author was present throughout the exhibition to explain the project, observe visitors’ reactions, and attend to their possible questions.

Based on the first author’s observation and conversations with the visitors, a typical behaviour around the artefact changed between:

- Restlessly checking if the calculation is updated/done or if it is correct.
- Being distant from the technology due to its stasis.

While these two seem to mean opposite of each other, some technical details such as the switching cycle of ARC updating the result (T_C and T_M) and the way the result is shown could influence the interaction as it is reported by Odom et al. [16]. In their work, the user engagement cycle of *Slow Game* has been adjusted to provide “dynamic, yet balanced quality of experience” (p. 77).

Another interesting remark was that the use of technology rather alienated the algae. One of the visitors commented that the black enclosure of the algal relay (which was designed to block the light from outside) portrays the algae as something more than just a living organism but instead a brain of the computer. The inclusion of the technology and how it was displayed have projected a certain character on algae.

The conversation with design researchers sometimes developed to the topic of care ethics, such as “whether this is good for algae”. Although care ethics has not been a focus of this project, and the prototype is not built from the perspective of better hospitality of algae culture, the emergence of this topic in discussions surfaced the inevitability of care relations in more-than-human design projects.

Even though the prototype was not tested in the intended way due to technical issues (i.e., leakage), presenting the concept through an engineered prototype has nevertheless evoked discussions on various topics: imagining possible interactions, perception of algae through technology and care.

4 FUTURE WORK

Based on the insights from the design process and the exhibition, we discuss the points to be addressed in future work.

One of the most addressed concerns during the exhibition, the duration of the calculation process, may have motivated from the audience’s perspectives on user experience or their interest in the biology of algae. The dominant anthropocentric value on productivity in modern society may have led to a desire to hasten the calculation process, however, this conflicts with the longer time required for the growth of algae. The prototype effectively demonstrated the coexistence of competing human and nonhuman temporalities and indicated its potential in prompting thoughts related to algae temporality.

Our design enabled the experience of only one kind of temporality, specifically the growth of algae. However, during our research and development process, we became aware of the existence of various temporalities even within one species of algae. For example, the life cycle of a micro-algal cell is significantly shorter than the

human lifespan. By focusing on this multiplicity and relativity of natural temporalities in future designs, we can incorporate broader expressions in our design and assist both designers and audiences in recognizing the complexity of time and temporality that surrounds us.

Conducting longitudinal observations of user interactions with the prototype could provide deeper insights into the tensions between human and nonhuman temporalities. In order to conduct these observations effectively, it is essential to give deeper consideration to the materials used for the algae reservoirs to prevent challenges in long-term use such as leakage. Furthermore, biological considerations on the energy conservation of algal activities are crucial to ensure the habitability of the prototype for micro-algae. Such considerations lead to the establishment of a care relationship between human and nonhuman, which we hope to further explore.

5 CONCLUDING REMARKS

The exploration of conceptualizing time and temporality beyond clock time is an emerging topic in HCI. While this perspective has the potential to broaden our perspectives in every day life by slowing down and bring the opportunity for reflection, the further design practices are needed to advance this topic. In light of growing attention to more-than-human design, which aims to reconnect humans with nature, design research focusing on nonhuman temporalities using tangible artefacts is valuable.

With this in mind, I have designed showroom artefact, ARC – Algal Relay Computer: a computer that incorporates nonhuman temporality, to evoke discussions regarding how we could incorporate more-than-human temporalities in our design practices. Despite the functional limitations of the prototype, the exhibition has brought discussions on many related topics to unpack, which means that it has served the purpose as a conversation opener. The starting point of this project, the slowness, was revealed to be only a limited interpretation of the complex, diverse more-than-human temporalities that coexist.

Through analysing the receptions and feedback, we have drawn a few reflection points to be considered deeper in future work:

- The competition between human and nonhuman temporalities that surfaces through the technological portrait.
- The multiplicity and relativity of temporalities that are worth addressing through artefacts.
- Material consideration that allows long-term interactions and subsequently establishes care relationships between human and nonhuman entities.

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